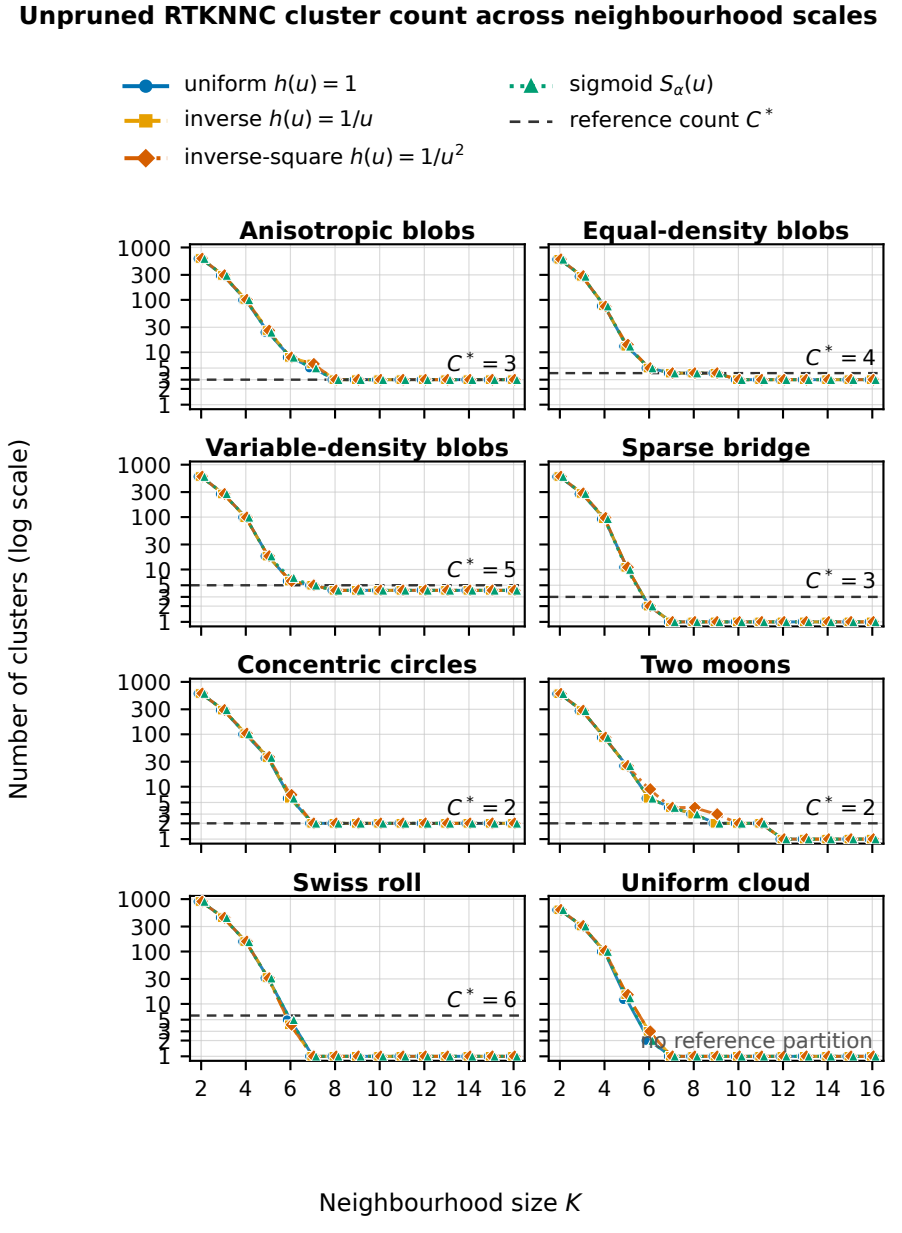
Unpruned RTKNNC cluster count across neighbourhood scales
uniform $h(u) = 1$
inverse $h(u) = 1/u$
inverse-square $h(u) = 1/u^2$
sigmoid $S_\alpha(u)$
reference count C^*
Anisotropic blobs
$C^* = 3$
Equal-density blobs
$C^* = 4$
Variable-density blobs
$C^* = 5$
Sparse bridge
$C^* = 3$
Concentric circles
$C^* = 2$
Two moons
$C^* = 2$
Swiss roll
$C^* = 6$
Uniform cloud
no reference partition
1000
300
100
30
10
5
3
2
1
2
4
6
8
10
12
14
16
Number of clusters (log scale)
Neighbourhood size K